



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 3310

Digital Fundamentals and System Design

Wenchao Miao



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Digital Fundamentals and System Design

Course Introduction

Wenchao Miao

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Course website	Canvas (Syllabus)
Instructor:	Wenchao Miao (eeuwenmiao@ust.hk) Office: 2411 Tel: 2358 5004
Teaching Assoc:	Raymundo Saijong Tang Tang (eeraytan@ust.hk) Rm 2395
TAs:	LI Shangkun, LI Zeyu, SHAO Kunming, WEN Chaozheng
Textbook:	Digital Fundamentals, Global Edition, 11 th Ed by Thomas L Floyd
Reference:	Digital Design, Global Edition, 6 th Ed by M. Morris Mano & Michael D. Ciletti

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Lectures: MON (13:30 – 14:50)
FRI (09:00 – 10:20)

Office Hours: MON (15:00 – 16:00)
FRI (10:30 – 11:30)

Tutorials: FRI (15:30 – 16:20)
FRI (16:30 – 17:20)

Labs: TUE (12:00 – 13:50)
TUE (15:00 – 16:50)

Lectures, Tutorials, and Labs will be delivered face-to-face until further notice.

Digital Fundamentals

ELEVENTH EDITION

GLOBAL EDITION

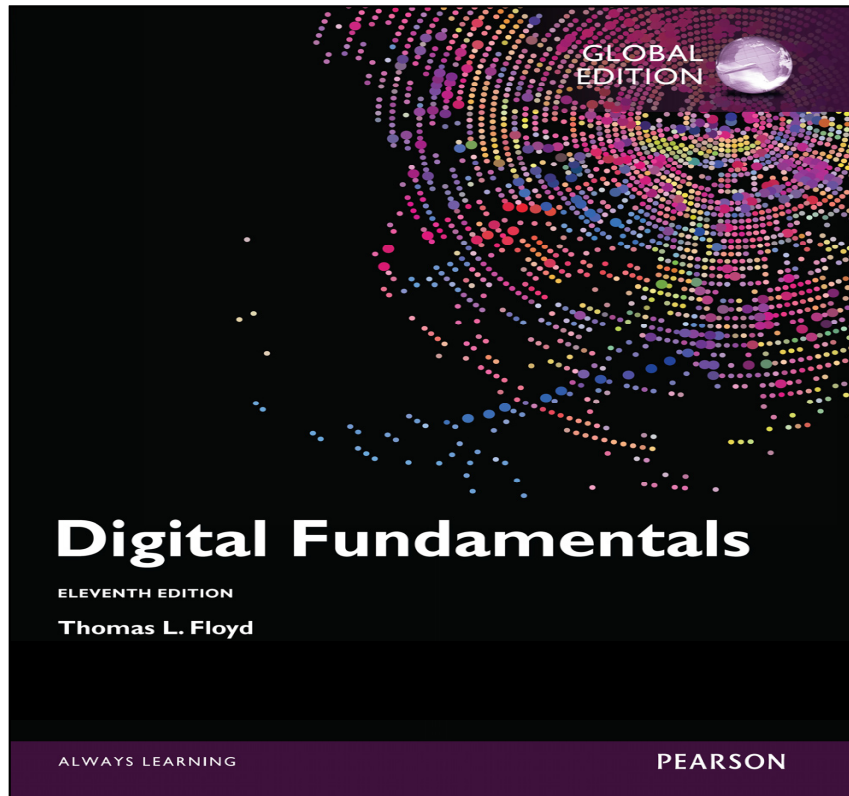
Course textbook

Strongly recommend to buy a copy

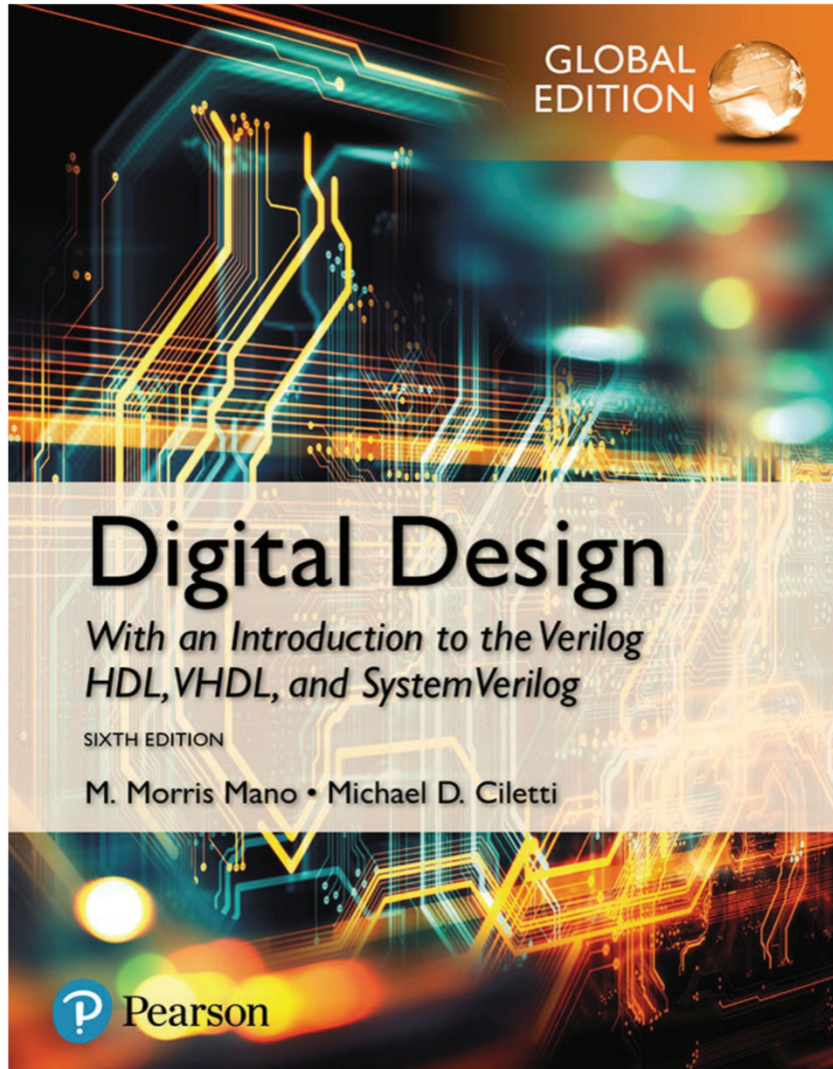
Title: *Digital Fundamentals,
Global Edition, 11th Edition*

Author: Thomas L Floyd

Print Book ISBN: 9781292075983



Reference book



Title: *Digital Design,
Global Edition, 6th Ed*

Author: M. Morris Mano &
Michael D. Ciletti

ISBN: 9781292231167

Library has the 4th Edition!

Course Intended Learning Outcomes

1. Describe and analyze sequential logic circuits
2. Design, model, and simulate sequential logic circuits using register-transfer level (RTL) design abstractions and hardware description languages (HDL)
3. Design, implement, and test sequential circuits and systems using field-programmable gate arrays (FPGAs)
4. Develop a model engineering system following a hierarchical design principle

Course Content

Week 1 – 5 Combinational Logic (Review Only)

Introductory Concepts, Numbers, Adder/Subtractor, Comparator, Boolean Expressions, Sum-of-products, product-of-sums, 1' s and 2' s complements, parity bit, error detection, logic blocks (decoders, encoders, multiplexers, demultiplexers)

Week 6 – 13 Sequential Logic (Main Part)

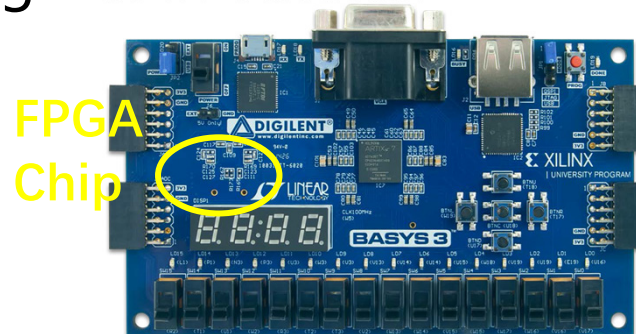
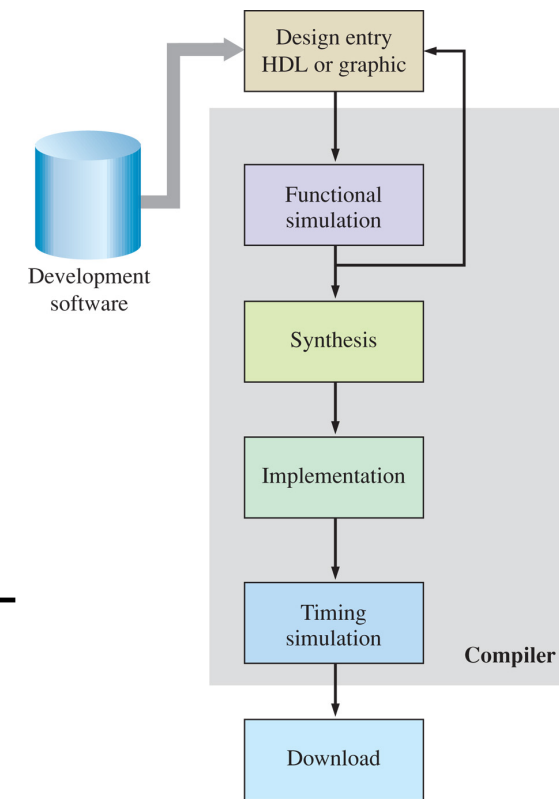
Combinational vs sequential, Clock vs enable, latches, flip-flops and registers, the concept of State and State diagram, Finite State Machine design, implementation, and analysis, counters, register transfer level design, control path and data path in the system, design automation flow

NOTE: To understand sequential logic circuits better, we will first briefly review combinational logic circuits

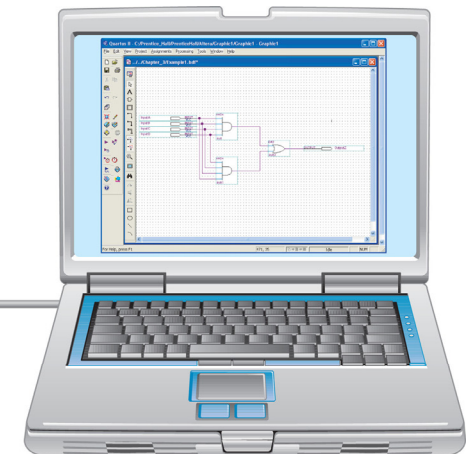
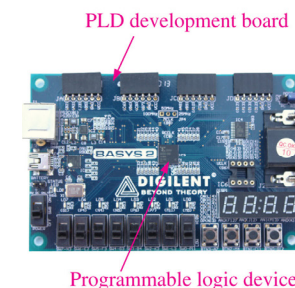
Midterm: OCT 20th (confirmed)

Labs & Project

- Total 6 Labs and 1 project all to be implemented on a ready-to-use digital circuit development platform (BASYS3 board)
- BASYS3 contains latest Field Programmable Gate Array (FPGA) chip from Xilinx (Artix®-7 from Xilinx®)
- You will use a hardware description language (HDL) known as VHDL to design and simulate your digital circuit on a development software platform (VIVADO)
- You will then download your program (digital circuit) to FPGA and then test on BASYS3 board
- Further details will be given in Tutorial



(Photo courtesy of Digilent, Inc.)



Labs & Project

Lab starts at Week 4

Week 4	Lab 1: Introduction to Vivado & FPGA Basys 3 board
Week 5	Lab 2: XOR Gate Array and Full Adder
Week 6	No Lab
Week 7	Lab 3: 4-bit Adder
Week 8	Lab 4: 4-bit Adder/Subtractor
Week 9	Lab 5: 4-bit binary up counter
Week 10&11	Lab 6: 4-bit Adder/Subtractor with 7-segment displays
Week 12	Project: 4-bit Overlapping Sequence detector
Week 13	Project demo [NOV 25]

Tutorials

- Tutorials and Lectures are independent
 - Lectures: you will learn fundamental concepts
 - Tutorials: you will learn VHDL, hardware description language to implement your digital circuits using FPGA
- Tutorials will also introduce each Lab a week before
- Tutorials will also involve solving practice problems

Course Grading Policy

- Midterm Exam 25%
- Final Exam 35%
- 6 Labs 25%
- Project 15%
- Homework will be graded and taken into consideration when further reference is needed about your performance
- This course will be assessed using criterion-referencing and grades will not be assigned using a curve. Details are provided in *Canvas -> Lecture Schedule -> ELEC3310 OBE (2025)*.
- The use of Generative AI in homework assignments, laboratory experiments and the project is permitted with proper acknowledgement.

ELEC3310

- First time being offered in Fall 2020
- Upgraded from ELEC2200 to make the course more relevant to today modern industry needs
- Updated course content
 - ✓ More emphasis on **sequential logic** and less on combinational logic
 - ✓ Updated lecture slides
 - ✓ More emphasis on **programmable ICs (FPGAs)** and less on fixed-function ICs (Integrated circuits)
 - ✓ Updated Labs using **BASYS 3 board (FPGA)**
 - ✓ More emphasis on **Project** component



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Thank you for listening !

